



The Impact of Information Sharing on Organ Transplantation: A Simulation Model

Zahra Gharibi and Michael Hahsler

Department of Engineering Management, Information, and Systems

INFORMS Annual Meeting 2018

Phoenix, AZ

World Changers
Shaped Here



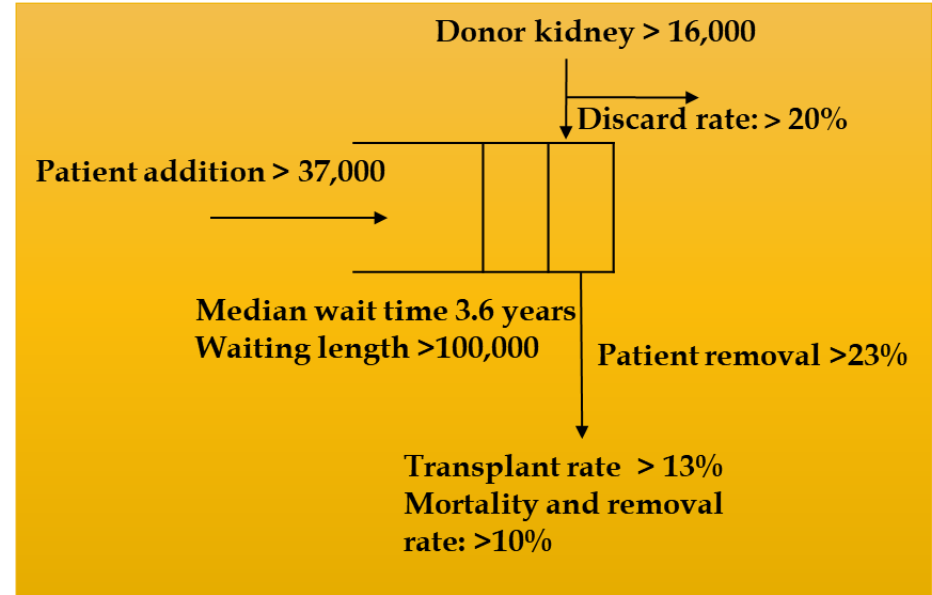
SMU®

Motivation

Key Question: Why Is The Kidney Discard Rate in the US Greater Than 20 Percent?

Donor Kidneys are discarded because of

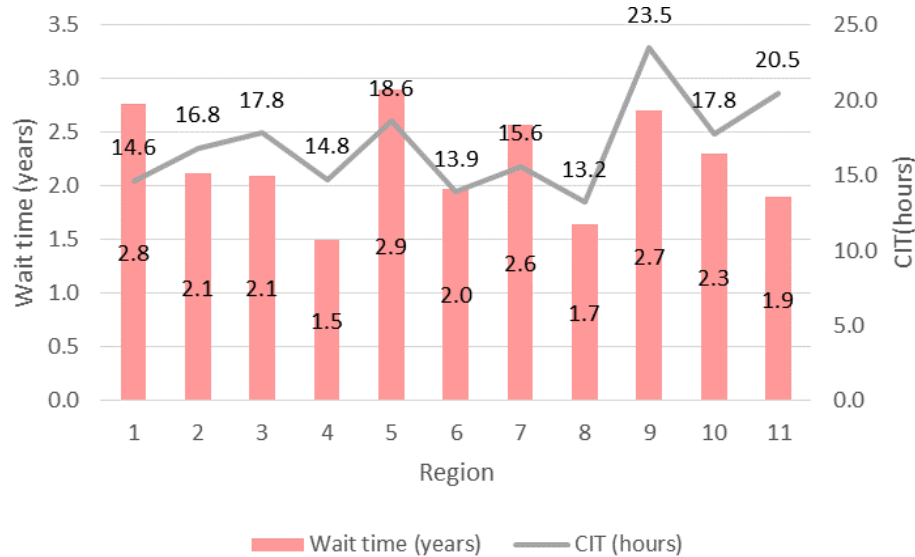
- ❑ Medical concerns
- ❑ Kidney allocation process
 - Time to find patient (CIT)
 - Early donor kidney rejection
 - Conditions of Participation (CoP)



Data sources: *United Network for Organ Sharing (UNOS)*

Motivation

Regional Differences in the US



Data sources: *United Network for Organ Sharing (UNOS) and Scientific Registry of Transplant Recipients (SRTR)*

Goal

Increase Donor Kidney Utilization

- Decision maker (patient & surgeon):

- Reconsidering decision criteria for acceptance/rejection of donor kidney
- Listing decision (Multiple listing)

- Social planner (OPTN & CMS):

- Changing kidney allocation system (KAS)
- Changes in Organ Procurement and Transplantation Network (OPTN; currently performed by UNOS) and CMS evaluation metrics
- Using information technology such as Unet to improve efficiency.

Information sharing



Literature Review

Kidney acceptance decision making

Ahn and Hornberger (1996), Howard (2002), Alagoz et al. (2004)

Allocation process and queueing models

Zenios et al. (2000), Su and Zenios (2004), Su and Zenios (2005), Su and Zenios (2006)

Simulation

Ruth et al. (1985)

Regional variations

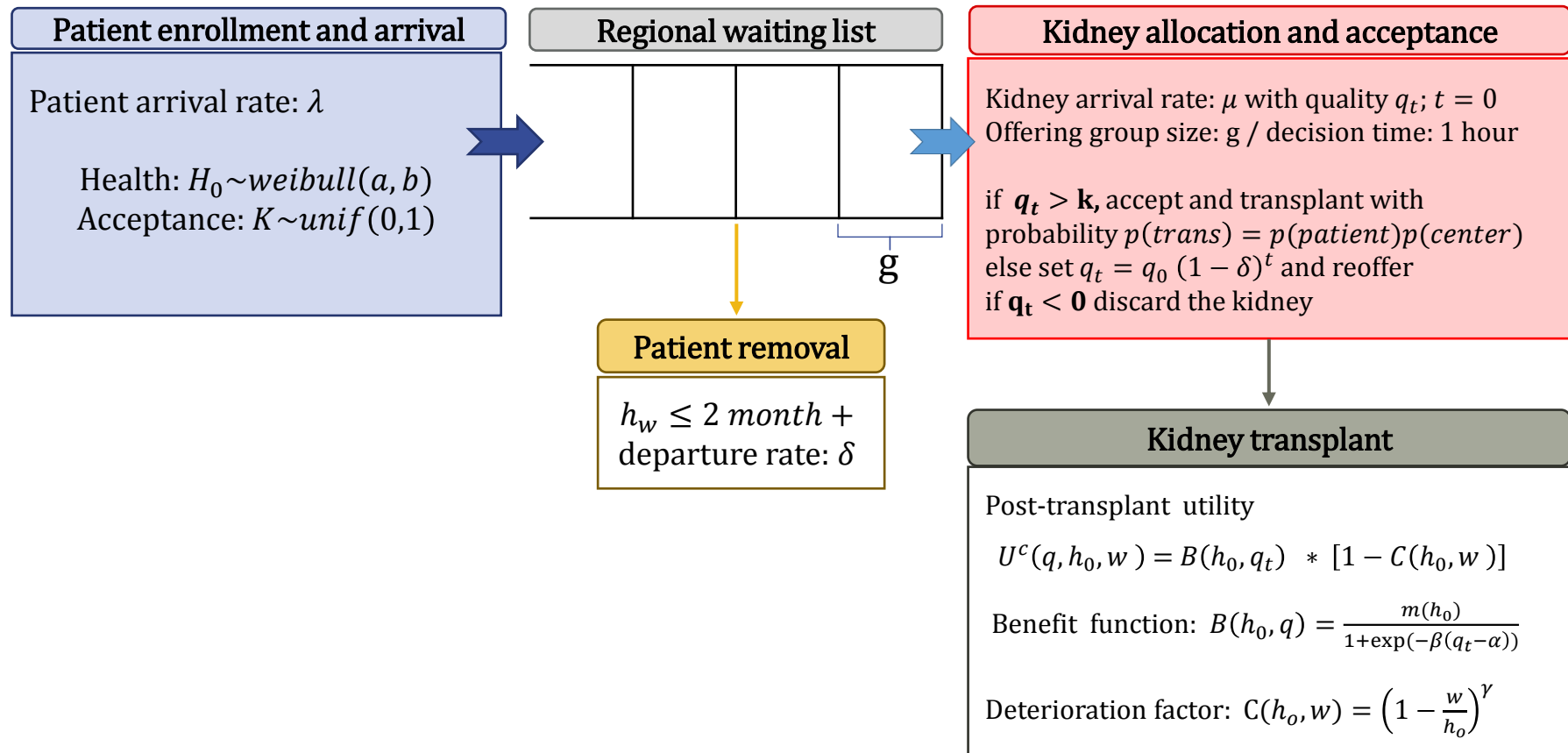
Bertsimas et al. (2013), Koizumi et al. (2015)

Method and Questions

Simulation optimization model to evaluate the impact of information sharing on organ transplantation.

1. Information sharing from OPTN to patient and surgeon.
 - What is the optimal **kidney acceptance policy** depending on region?
 - How to find the optimal **region selection policy** under a set of budget, distance and performance constraints?
2. Information sharing from patient/surgeon with OPTN
 - How does **information sharing** improve the efficiency of the kidney allocation system and enhance social welfare utility?

Simulation Model & Assumptions



Method and Questions

Simulation optimization model to evaluate the impact of information sharing on organ transplantation.

1. Information sharing from OPTN to patient and surgeon.
 - What is the optimal **kidney acceptance policy** depending on region?
 - How to find the optimal **region selection policy** under a set of budget, distance and performance constraints?
2. Information sharing from patient/surgeon with OPTN
 - How does **information sharing** improve the efficiency of the kidney allocation system and enhance social welfare utility?

Optimal Kidney Acceptance Threshold

Simulation

$$\max_k E[U_i]$$

$$E[U_i] = f(i, k)$$

k ... kidney acceptance threshold

i ... region

Estimation of Arrival Rates

Recipient with blood type A

Region	Donor A (94%)	Donor O (2%)	Kidney μ	Patient λ
1	118	143	228	537
2	387	473	746	1769
3	461	625	892	1680
4	303	421	586	1154
5	452	636	875	1910
6	132	165	255	377
7	239	300	461	1008
8	226	296	437	688
9	114	146	220	735
10	284	349	548	1040
11	304	423	588	1080

Data sources: *United Network for Organ Sharing (UNOS) and Scientific Registry of Transplant Recipients (SRTR)*

Optimal Kidney Acceptance Threshold by Region

Blood type	h_o	a	b	θ	α	β	γ	t	δ	g	$P(trans)$
A	1(yr)	8	2	5	0.4	8	0.5	1 (hr)	0.05	5	0.8



Region	μ	k_i^*	q_t	$U_i(k_i^*)$	n_{reject}
1	228	0.6	0.76	9.1	17
2	746	0.85	0.95	13.2	27
3	892	0.85	0.93	13.6	24
4	588	0.8	0.9	12.8	24
5	875	0.85	0.92	13.5	24
6	255	0.65	0.76	9.7	19
7	461	0.75	0.8	12.0	21
8	436	0.75	0.84	11.9	21
9	220	0.6	0.67	8.9	17
10	548	0.8	0.92	12.7	25
11	588	0.8	0.92	12.8	24

Method and Questions

- **Simulation optimization model** to evaluate the impact of information sharing on organ transplantation.
 1. Information sharing from OPTN to patient and surgeon.
 - What is the optimal **kidney acceptance policy** depending on region?
 - How to find the optimal **region selection policy** under a set of budget, distance and performance constraints?
 2. Information sharing from patient/surgeon with OPTN
 - How does **information sharing** improve the efficiency of the kidney allocation system and enhance social welfare utility?

Optimal Decision Without Constraints

Blood type	h_o	a	b	θ	α	β	γ	t	δ	g	$P(trans)$
A	1(yr)	8	2	5	0.4	8	0.5	1 (hr)	0.05	5	0.8



Region	μ	k_i^*	q_t	$U_i(k_i^*)$	n_{reject}
1	228	0.6	0.76	9.1	17
2	746	0.85	0.95	13.2	27
3	892	0.85	0.93	13.6	24
4	588	0.8	0.9	12.8	24
5	875	0.85	0.92	13.5	24
6	255	0.65	0.76	9.7	19
7	461	0.75	0.8	12.0	21
8	436	0.75	0.84	11.9	21
9	220	0.6	0.67	8.9	17
10	548	0.8	0.92	12.7	25
11	588	0.8	0.92	12.8	24

Region Selection

Simulation

$$\max_{r,k} E[U]$$

$$E[U] = f(r, k) = \sum_{i \in I} \mathbf{U}_i(\mathbf{k}_i^*) r_i$$

s. t.

$$\mathbf{r} = \{0, 1\}^{11}$$

r_i : 1 if region i is selected, 0 otherwise

$$\sum_{i \in I} r_i = 1$$

Single region constraint

$$\sum_{i \in I} c_i r_i \leq B$$

Pre transplant budget contr. (evaluation, travel + living cost)

$$\sum_{i \in I} d_i r_i \leq D$$

Distance constraint

$$\sum_{i \in I} p_i r_i \geq P$$

Performance constr. (5-year post-transplant graft survival)

Region Selection

Patient with blood type A living in Region 6

Constraints: Budget <\$15,000; Performance>75%, Distance <1,500 miles

Region	Monthly cost of living	Expected wait time (yrs)	Number of evaluation	Evaluation cost	5 year survival rate	Air distance(mile)	$U_i(k_i^*)(yr)$
1	\$2,454	2.4	4	\$6,381	73%	2,200	9.1
2	\$1,879	1.8	3	\$4,614	70%	1,904	13.2
3	\$1,637	1.9	3	\$4,541	76%	1,706	13.6
4	\$2,328	1.4	2	\$3,166	75%	1,160	12.8
5	\$2,130	2.5	5	\$7,815	79%	374	13.5
6	\$2,413	1.6	3	0\$	83%	0	9.7
7	\$2,686	2.3	4	\$6,474	74%	670	12.0
8	\$2,869	1.4	2	\$3,274	76%	569	11.9
9	\$2,705	2.2	4	\$6,482	69%	1,195	8.9
10	\$2,005	1.8	3	\$4,651	72%	565	12.7
11	\$2,706	1.4	2	\$3,241	72%	941	12.8

Set of feasible regions : {R6, R5, R4, R8}

Proposed Approach for Multi-Listing

Simulation

$$\max_{\mathbf{r}, \mathbf{k}} E[U]$$

$$E[U] = f(\mathbf{r}, \mathbf{k}) = U_{\mathbf{r}, \mathbf{k}}$$

s. t.

$$\mathbf{r} = \{0, 1\}^{11}$$

r_i : 1 if region i is selected, 0 otherwise

~~$$\sum_{i \in I} r_i = 1$$~~

~~Single region constraint~~

$$\sum_{i \in I} c_i r_i \leq B$$

Pre transplant budget contr. (evaluation, travel + living cost)

$$\sum_{i \in I} d_i r_i \leq D$$

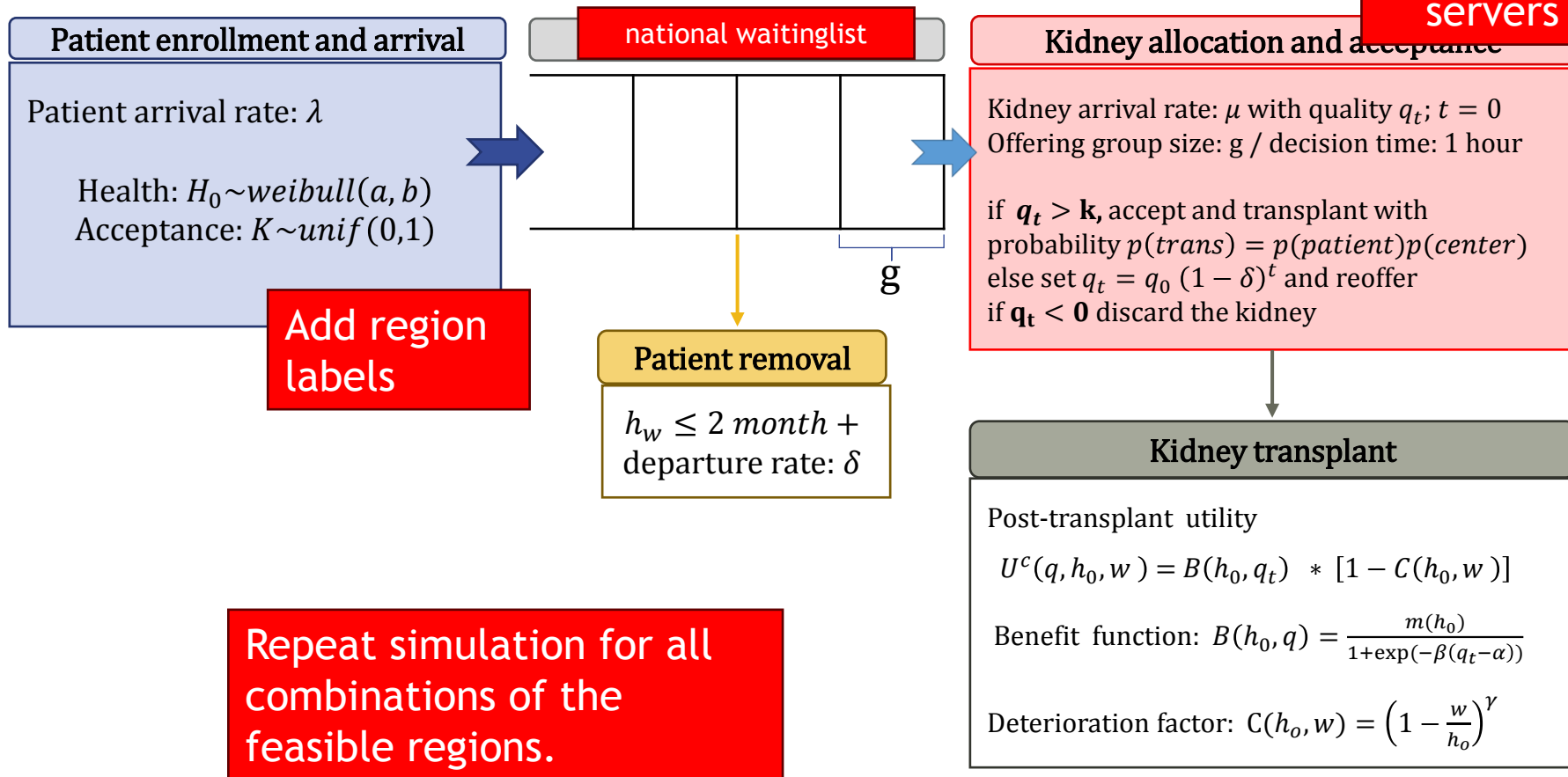
Distance constraint

$$\sum_{i \in I} p_i r_i \geq P$$

Performance constr. (5-year post-transplant graft survival)

Simulation Model for Multi-Listing

11 regional
servers



Method and Questions

- **Simulation optimization model** to evaluate the impact of information sharing on organ transplantation.
 1. Information sharing from OPTN to patient and surgeon.
 - What is the optimal **kidney acceptance policy** depending on region?
 - How to find the optimal **region selection policy** under a set of budget, distance and performance constraints?
 2. Information sharing from patient/surgeon with OPTN
 - How does **information sharing** improve the efficiency of the kidney allocation system and enhance social welfare utility?

The Effect of Information Sharing on Patient Transplant Outcomes

Current situation

- Kidney is offered to a set of $g = 5$ patients which have 1 hour to decide.
- Kidney is transplanted for each of the g patient with $p(trans) = p(patient)p(center)$
- If the kidney is not accepted it is re-offered to the next group.

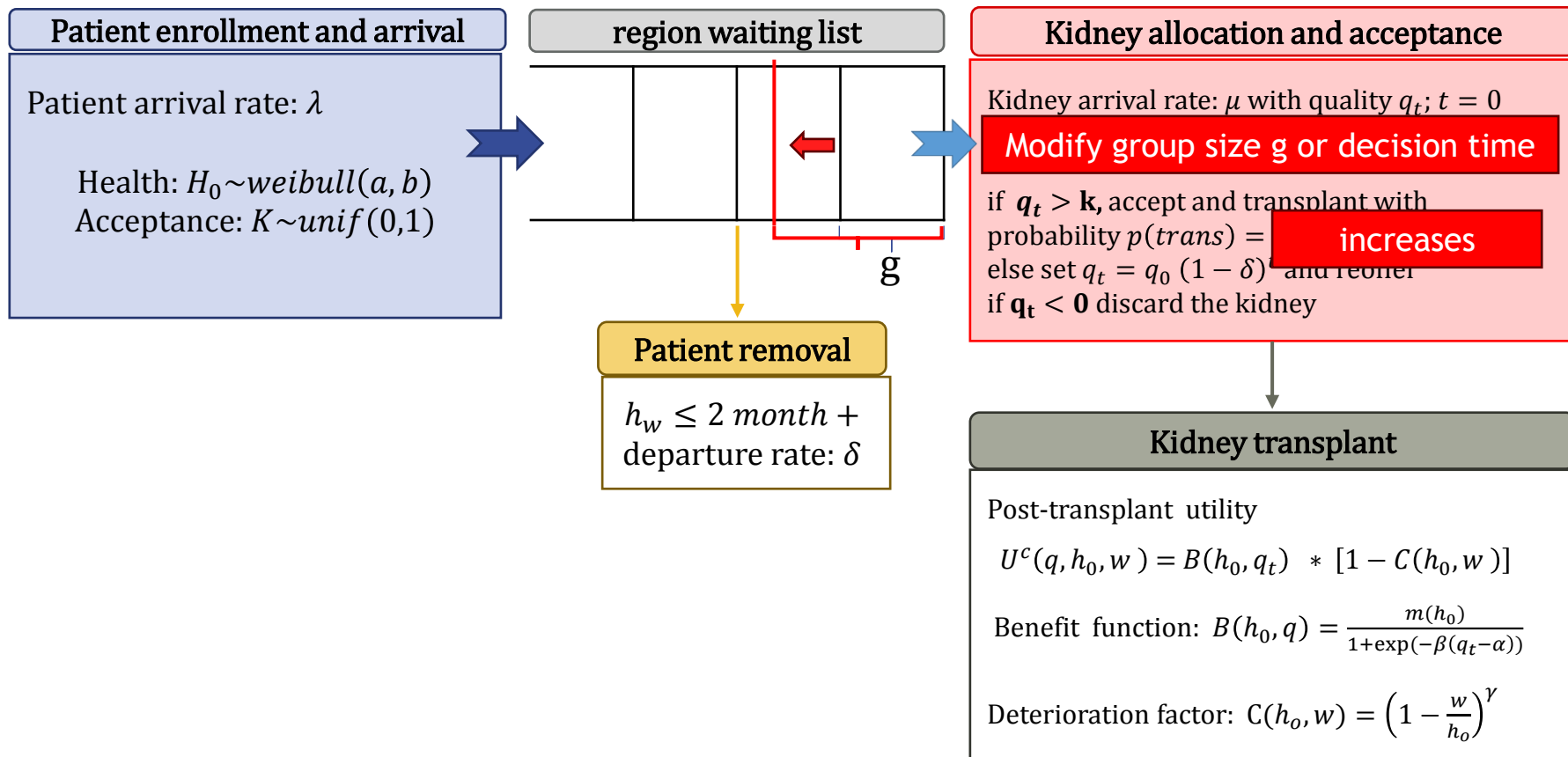
Under perfect information OPTN can use for allocation

- All patient's kidney quality thresholds.
- Patient's availability.
- Surgeon and center availability.

OPTN can identify the perfect candidate directly and CIT is minimized.
This can be expressed as a group size of $g = \infty$.

Effect of information sharing can be modeled as an increase of g , a decrease of decision time and an increase of $p(trans)$.

Simulation Model for Information Sharing



The Effect of Information Sharing on Patient Post Transplant Utility Based on Region 6 Kidneys Supply and Demands

Group size g	Waitlist position (mean)	Waitlist position (max)	Average q	Average utility (year)	Total utility (year)
5	6	45	0.66	11.03	284.48
10	8	80	0.62	10.99	312.72
20	10	119	0.6	10.9	321.17
100	22	344	0.57	10.82	331.22
∞	58	1,383	0.55	10.80	339

Kidney Utilization, Waitlist Removal, and Kidney Transplant Rates In Region 6

Group size	Kidney utilization rate	Kidney discard rate	Waitlist removal rate	Kidney transplant rate
5	85%	15%	8.9%	17%
10	90.9%	9.1%	8.3%	20%
20	94.3%	5.7%	7.9%	21.5%
100	98.5%	1.5%	7.3%	23.8%
∞	100%	0%	7.1%	25.2%

Concluding Remarks

- High organ discard rates are a known problem.
- UNOS encourages the development of modern technology such as mobile devices to facilitate organ allocation.
- The presented simulation model shows what the impact of different information strategies are.
- Challenge: Incentive system for information sharing.

Questions?

Contact Information

Michael Hahsler

mhahsler@lyle.smu.edu

